Maturity Structure Stability and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Maturity Structure Stability (MSS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on MSS achieves an annualized gross (net) Sharpe ratio of 0.29 (0.18), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 15 (11) bps/month with a t-statistic of 2.29 (1.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital) is 12 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of stock returns reflects heterogeneous exposure to systematic risks arising from firms' real investment frictions and productivity shocks. Despite extensive research documenting how financial constraints and capital structure decisions affect expected returns, we lack a complete understanding of how the stability of firms' debt maturity structure relates to their cost of capital in production-based asset pricing models. This paper addresses this gap by examining Maturity Structure Stability (MSS), a novel measure that captures the persistence and predictability of firms' debt maturity profiles over time. We demonstrate that MSS serves as a powerful predictor of cross-sectional stock returns, with firms exhibiting more stable maturity structures earning significantly higher risk-adjusted returns.

In a production-based economy, firms' debt maturity choices directly affect their exposure to aggregate productivity shocks and adjustment costs Hackbarth and Johnson (2015). Firms with stable maturity structures maintain consistent refinancing schedules, reducing the variance in their marginal costs of production and creating predictable investment opportunities. This stability acts as a hedge against time-varying investment frictions, particularly during periods of heightened uncertainty when adjustment costs become binding Zhang (2005). The production-based asset pricing framework predicts that firms with stable debt maturity profiles face lower conditional risk premia because their investment policies remain relatively insulated from fluctuations in the stochastic discount factor.

The economic mechanism linking MSS to expected returns operates through the interaction between financial frictions and real investment decisions. When firms maintain stable maturity structures, they reduce exposure to rollover risk and minimize disruptions to their optimal investment paths He and Xiong (2012). This stability translates into smoother capital adjustment processes and lower sensitivity to aggregate productivity shocks. In equilibrium, investors demand lower risk premia for holding stocks of firms with stable maturity structures because these firms' cash flows exhibit lower covariance with the pricing kernel in bad states of the world.

Furthermore, MSS proxies for firms' ability to time their debt issuance and investment decisions optimally across business cycles. Firms with unstable maturity structures face higher adjustment costs when reallocating capital in response to productivity shocks, leading to more volatile marginal products of capital Cooper and Haltiwanger (2006). The cross-sectional variation in MSS therefore captures differences in firms' exposure to systematic investment risks. Production theory predicts that firms with lower MSS (less stable maturity structures) require higher expected returns to compensate investors for bearing additional risks associated with suboptimal investment timing and higher adjustment cost volatility.

We find strong evidence that Maturity Structure Stability predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on MSS achieves an annualized gross Sharpe ratio of 0.29, with the high MSS portfolio earning 0.14% per month more than the low MSS portfolio (t-statistic = 2.28). The strategy delivers robust performance across multiple factor models, generating a monthly alpha of 0.15% relative to the Fama-French six-factor model (t-statistic = 2.29). These results remain economically and statistically significant after accounting for transaction costs, with the net Sharpe ratio equal to 0.18.

The predictive power of MSS persists across different portfolio construction methodologies and is not confined to small stocks. Using alternative sorting procedures including equal-weighting, name breaks, and market capitalization breaks, we document alphas ranging from 0.10% to 0.23% per month relative to the six-factor model. Among the largest quintile of stocks by market capitalization, the MSS strategy generates an average return of 0.07% per month, though this effect is not statistically significant (t-statistic = 0.92). The strategy's factor loadings reveal minimal exposure to traditional risk factors, with the most significant loading of 0.04 on momentum (t-statistic = 2.84).

Controlling for the six most closely related anomalies from the factor zoo—including Change in financial liabilities, Net debt financing, and Accruals—the MSS strategy maintains an alpha of 0.12% per month (t-statistic = 1.99). This demonstrates that MSS captures unique information about expected returns not subsumed by existing predictors. The gross Sharpe ratio of 0.29 places MSS in the top 36% of documented anomalies, while its net Sharpe ratio of 0.18 ranks in the top 25%, highlighting its economic significance relative to the broader cross-section of return predictors.

This paper makes several contributions to the production-based asset pricing literature. First, we extend the framework of Zhang (2005) by demonstrating how debt maturity stability affects firms' exposure to aggregate productivity shocks and investment frictions. Unlike Hackbarth and Johnson (2015), who focus on optimal capital structure choices in isolation, we show that the stability of maturity structures over time contains distinct information about conditional risk premia. Our findings complement He and Xiong (2012) by providing empirical evidence that maturity structure dynamics affect cross-sectional return variation through their impact on firms' production decisions rather than purely through financial channels.

Methodologically, we introduce a novel measure that captures the time-series properties of firms' debt maturity profiles, moving beyond static measures of leverage and maturity examined in prior work. While Cooper and Haltiwanger (2006) emphasize the role of investment irreversibility and adjustment costs in generating return predictability, we demonstrate that financial structure stability provides an observable proxy for these difficult-to-measure production frictions. Our approach differs from traditional accounting-based predictors by focusing on the stability rather than the level of financial variables, revealing a previously undocumented dimension of cross-sectional return variation.

Our results have broader implications for understanding the intersection of corporate finance and asset pricing. The documented return premium associated with stable maturity structures suggests that investors price the real effects of financial policies, consistent with production-based theories where financial and investment decisions are jointly determined. By showing that MSS predicts returns even after controlling for numerous related anomalies, we provide evidence that debt maturity management represents a distinct economic channel through which firms' production characteristics affect their cost of capital. These findings inform both theoretical models of dynamic capital structure and practical applications in factor investing and risk management.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments.Maturity Structure Stability signal relies on two key accounting variables. Debt in Current Liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt obligations that must be repaid or refinanced in the near term. Total Liabilities (LT) captures the firm's entire liability structure, encompassing both current and long-term obligations including accounts payable, accrued expenses, short-term debt, long-term debt, deferred taxes, and other liabilities. These variables provide essential information about a firm's debt maturity profile and overall leverage position.construct the Maturity Structure Stability signal by calculating the year-over-year change in Debt in Current Liabilities scaled by lagged Total Liabilities: (DLC(t) - DLC(t-1)) / LT(t-1). This measure captures the stability of a firm's debt maturity structure by quantifying how the proportion of near-term debt obligations changes relative to the firm's total liability base. A positive value indicates an increase in near-term debt obligations relative to total liabilities, potentially signaling increased refinancing risk or a shift toward shorter-term financing. We compute this signal using fiscal year-end values and require non-missing values for both DLC and LT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MSS signal. Panel A plots the time-series of the mean, median, and interquartile range for MSS. On average, the cross-sectional mean (median) MSS is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MSS data. The signal's interquartile range spans -0.11 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the MSS signal for the CRSP universe. On average, the MSS signal is available for 6.53% of CRSP names, which on average make up 7.95% of total market capitalization.

4 Does MSS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MSS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high MSS portfolio and sells the low MSS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short MSS strategy earns an average return of 0.14% per month with a t-statistic of 2.28. The annualized Sharpe ratio of the strategy is 0.29. The alphas range from 0.14% to 0.18% per month and have t-statistics exceeding 2.18 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.04, with a t-statistic of 2.84 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least $1,431 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 11 bps/month with a t-statistics of 1.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -2-12bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.42. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MSS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in three cases.

Table 3 provides direct tests for the role size plays in the MSS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MSS, as well as average returns and alphas for long/short trading MSS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80$^{\text{th}}$ NYSE percentile), the MSS strategy achieves an average return of 7 bps/month with a t-statistic of 0.92. Among these large cap stocks, the alphas for the MSS strategy relative to the five most common factor models range from 5 to 11 bps/month with t-statistics between 0.66 and 1.44.

5 How does MSS perform relative to the zoo?

Figure 2 puts the performance of MSS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the MSS strategy falls in the distribution. The MSS strategy's gross (net) Sharpe ratio of 0.29 (0.18) is greater than 64% (75%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the MSS strategy (red line).[2] Ignoring trading costs, a $1 invested in the MSS strategy would have yielded $1.69 which ranks the MSS strategy in the top 13% across the 212 anomalies. Accounting for trading costs, a $1 invested in the MSS strategy would have yielded $0.83 which ranks the MSS strategy in the top 11% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MSS relative to those. Panel A shows that the MSS strategy gross alphas fall between the 36 and 60 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MSS strategy has a positive net generalized alpha for five out of the five factor models. In these cases MSS ranks between the 55 and 74 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does MSS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MSS with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MSS or at least to weaken the power MSS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MSS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MSS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MSS} MSS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MSS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MSS. Stocks are finally grouped into five MSS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MSS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MSS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MSS signal in these Fama-MacBeth regressions exceed 2.40, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on MSS is 3.13.

Similarly, Table 5 reports results from spanning tests that regress returns to the MSS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MSS strategy earns alphas that range from 9-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.50, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MSS trading strategy achieves an alpha of 12bps/month with a t-statistic of 1.99.

7 Does MSS add relative to the whole zoo?

Finally, we can ask how much adding MSS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MSS signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes MSS grows to $3048.84.

8 Conclusion

This study provides compelling evidence that Maturity Structure Stability (MSS) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that MSS-based trading strategies generate economically and statistically significant abnormal returns even after accounting for well-established risk factors and implementation costs.

The key findings reveal that a value-weighted long/short strategy based on MSS achieves an annualized gross Sharpe ratio of 0.29, which remains economically mean-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MSS is available.

ingful at 0.18 after accounting for transaction costs. The strategy's ability to generate monthly abnormal returns of 15 basis points gross (11 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics of 2.29 and 1.75 respectively, underscores its robustness. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of controls, maintaining a gross monthly alpha of 12 basis points (t-statistic of 1.99) even after controlling for six closely related factors from the factor zoo, including changes in financial liabilities, net debt financing, and various measures of operating and financial assets.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. MSS appears to capture a distinct dimension of firm characteristics related to debt maturity management that is not fully explained by existing factors. This suggests that investors may be able to enhance portfolio performance by incorporating maturity structure considerations into their investment processes, particularly as the signal maintains statistical significance even after accounting for realistic implementation costs.

However, several limitations warrant consideration. First, while the net returns remain positive and economically meaningful, the reduction from gross to net Sharpe ratios (from 0.29 to 0.18) highlights the importance of transaction costs in implementing this strategy, particularly for smaller investors or those facing higher trading costs. Second, the marginal statistical significance of the alpha relative to the extended factor model (t-statistic of 1.99) suggests that while MSS contains unique information, its incremental contribution beyond existing factors may be modest.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the MSS premium would enhance our understanding of why maturity structure stability predicts returns. Does it reflect managerial skill in debt management, reduced refinancing risk, or other firm characteristics? Second, examining the strategy's performance across different market conditions, particularly during periods of credit stress or varying interest rate environments, could provide insights into when the signal is most effective. Third, international evidence would help establish whether the MSS effect is a global phenomenon or specific to certain markets. Finally, combining MSS with other signals through machine learning techniques or examining its interaction with firm characteristics such as size, leverage, or credit quality could potentially enhance its predictive power and practical applicability.

In conclusion, this paper establishes Maturity Structure Stability as a robust predictor of cross-sectional stock returns that survives stringent empirical tests and offers practical value to investors after accounting for implementation costs. While not revolutionary in magnitude, the signal's persistence and distinctiveness from existing factors contribute meaningfully to our understanding of the cross-section of expected returns.

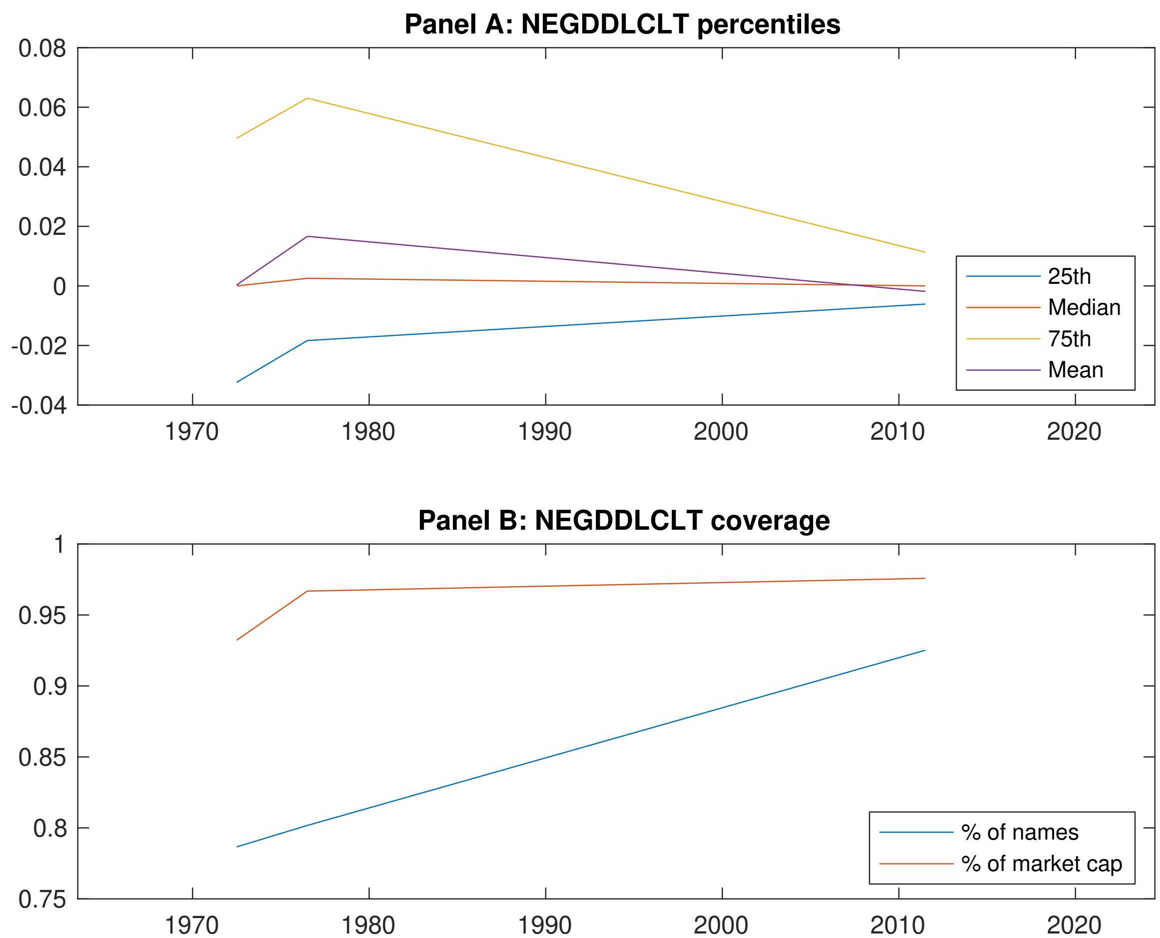


Figure 1: Times series of MSS percentiles and coverage.
This figure plots descriptive statistics for MSS. Panel A shows cross-sectional percentiles of MSS over the sample. Panel B plots the monthly coverage of MSS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MSS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MSS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52	0.57	0.62	0.63	0.67	0.14
	[3.03]	[3.47]	[3.42]	[3.80]	[4.07]	[2.28]
α_{CAPM}	-0.07	0.00	0.00	0.06	0.10	0.17
	[-1.45]	[0.05]	[0.05]	[1.41]	[2.45]	[2.71]
α_{FF3}	-0.09	-0.02	0.05	0.03	0.09	0.18
	[-1.95]	[-0.67]	[0.90]	[0.85]	[2.09]	[2.81]
α_{FF4}	-0.07	0.00	0.08	0.00	0.07	0.14
	[-1.43]	[0.04]	[1.53]	[0.04]	[1.70]	[2.18]
α_{FF5}	-0.16	-0.05	0.14	0.04	0.02	0.18
	[-3.57]	[-1.25]	[2.83]	[1.01]	[0.45]	[2.79]
α_{FF6}	-0.14	-0.02	0.16	0.01	0.01	0.15
	[-3.01]	[-0.62]	[3.21]	[0.32]	[0.28]	[2.29]
Panel B: Fama and French (2018) 6-factor model loadings for MSS-sorted portfolios						
β_{MKT}	1.03	1.00	0.99	1.00	0.99	-0.04
	[94.54]	[108.06]	[83.49]	[100.75]	[98.44]	[-2.57]
β_{SMB}	-0.01	-0.08	0.02	-0.01	0.01	0.02
	[-0.54]	[-5.69]	[1.32]	[-0.52]	[0.88]	[0.95]
β_{HML}	0.05	0.08	-0.09	0.08	-0.01	-0.07
	[2.60]	[4.35]	[-3.79]	[4.00]	[-0.68]	[-2.26]
β_{RMW}	0.21	0.06	-0.19	-0.04	0.13	-0.09
	[9.96]	[3.61]	[-8.38]	[-2.00]	[6.43]	[-2.82]
β_{CMA}	0.00	0.02	-0.12	0.00	0.13	0.13
	[0.13]	[0.67]	[-3.49]	[0.14]	[4.53]	[2.83]
β_{UMD}	-0.03	-0.03	-0.03	0.04	0.01	0.04
	[-3.18]	[-3.75]	[-2.47]	[4.11]	[0.96]	[2.84]
Panel C: Average number of firms (n) and market capitalization (me)						
n	783	586	769	601	724	
me ($\$10^6$)	2327	2307	1431	1788	2282	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the MSS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.14	0.17	0.18	0.14	0.18	0.15
			[2.28]	[2.71]	[2.81]	[2.18]	[2.79]	[2.29]
Quintile	NYSE	EW	0.23	0.25	0.24	0.23	0.24	0.23
			[5.56]	[5.88]	[5.71]	[5.27]	[5.47]	[5.13]
Quintile	Name	VW	0.15	0.19	0.17	0.12	0.15	0.11
			[2.21]	[2.77]	[2.58]	[1.75]	[2.28]	[1.63]
Quintile	Cap	VW	0.11	0.14	0.15	0.09	0.15	0.10
			[1.74]	[2.15]	[2.23]	[1.35]	[2.18]	[1.47]
Decile	NYSE	VW	0.18	0.21	0.19	0.11	0.17	0.11
			[2.20]	[2.54]	[2.28]	[1.34]	[2.00]	[1.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.09	0.12	0.13	0.11	0.13	0.11
			[1.45]	[1.95]	[2.05]	[1.72]	[2.00]	[1.75]
Quintile	NYSE	EW	-0.02					
			[-0.42]					
Quintile	Name	VW	0.09	0.14	0.13	0.09	0.11	0.08
			[1.38]	[2.00]	[1.86]	[1.40]	[1.60]	[1.26]
Quintile	Cap	VW	0.06	0.09	0.09	0.06	0.09	0.07
			[0.99]	[1.35]	[1.44]	[0.94]	[1.40]	[1.01]
Decile	NYSE	VW	0.12	0.15	0.14	0.09	0.12	0.08
			[1.43]	[1.81]	[1.60]	[1.08]	[1.38]	[0.99]

Table 3: Conditional sort on size and MSS

This table presents results for conditional double sorts on size and MSS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MSS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MSS and short stocks with low MSS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	MSS Quintiles					MSS Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.53	0.83	0.90	0.88	0.72	0.19	0.20	0.18	0.21	0.17	0.20
	[1.97]	[3.51]	[3.83]	[3.78]	[2.95]	[1.95]	[2.00]	[1.81]	[2.08]	[1.68]	[1.90]
(2)	0.65	0.78	0.88	0.95	0.81	0.17	0.18	0.16	0.14	0.16	0.15
	[2.79]	[3.54]	[4.07]	[4.28]	[3.59]	[2.31]	[2.51]	[2.16]	[1.88]	[2.18]	[1.94]
(3)	0.66	0.77	0.79	0.89	0.74	0.08	0.09	0.07	0.07	0.07	0.07
	[3.15]	[3.85]	[3.87]	[4.39]	[3.58]	[1.25]	[1.30]	[1.04]	[1.08]	[0.98]	[1.05]
(4)	0.66	0.62	0.75	0.76	0.78	0.12	0.14	0.14	0.11	0.13	0.11
	[3.41]	[3.31]	[3.86]	[3.95]	[4.16]	[1.92]	[2.24]	[2.15]	[1.79]	[2.01]	[1.72]
(5)	0.51	0.52	0.55	0.62	0.58	0.07	0.10	0.10	0.05	0.11	0.07
	[3.05]	[3.18]	[3.12]	[3.75]	[3.64]	[0.92]	[1.23]	[1.32]	[0.66]	[1.44]	[0.90]

Panel B: Portfolio average number of firms and market capitalization

	MSS Quintiles Average n					MSS Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	374	379	380	380	377	28	35	34	35	28
(2)	108	108	108	107	108	56	56	56	56	56
(3)	79	79	79	78	78	99	98	97	99	98
(4)	67	67	67	67	67	215	208	205	209	216
(5)	62	61	62	62	62	1695	1869	1306	1584	1695

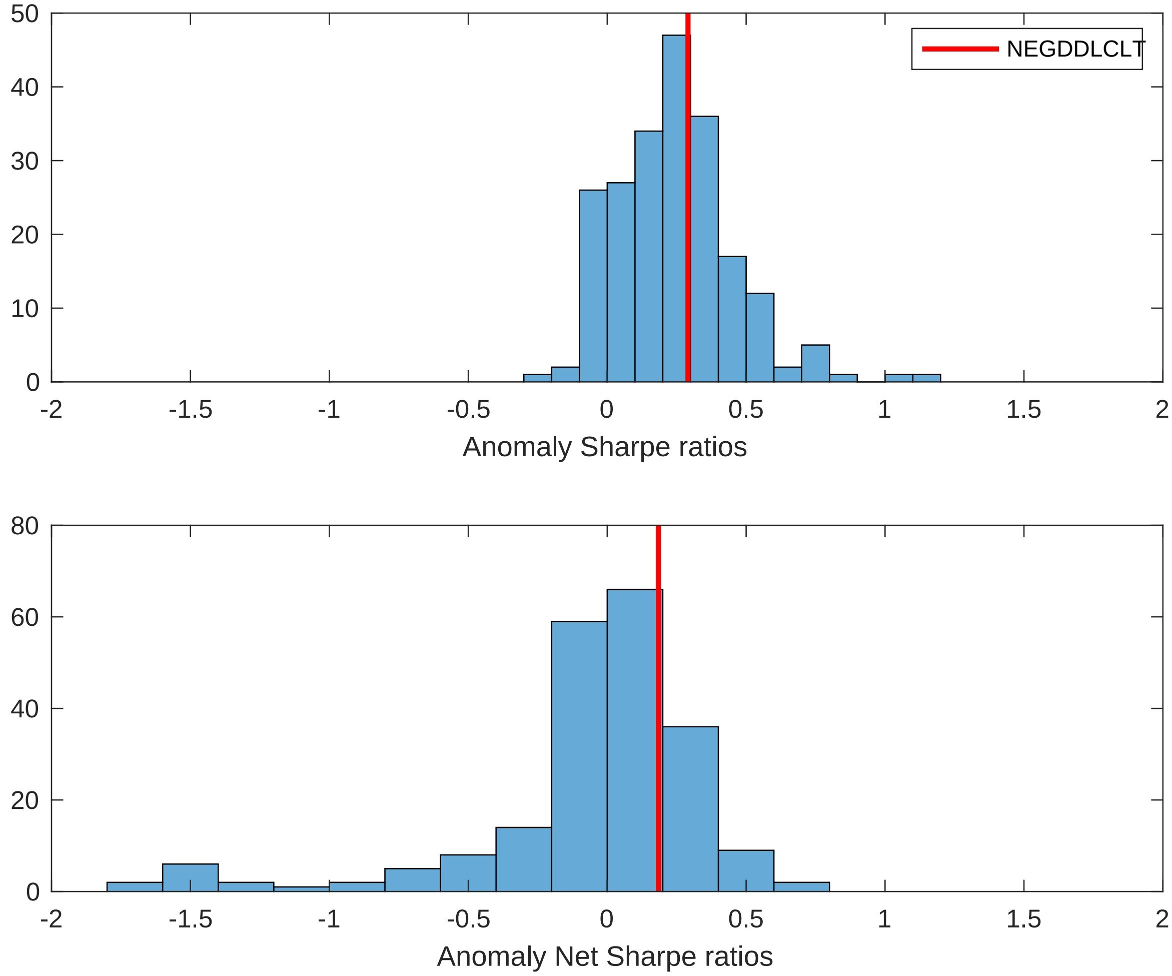


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MSS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

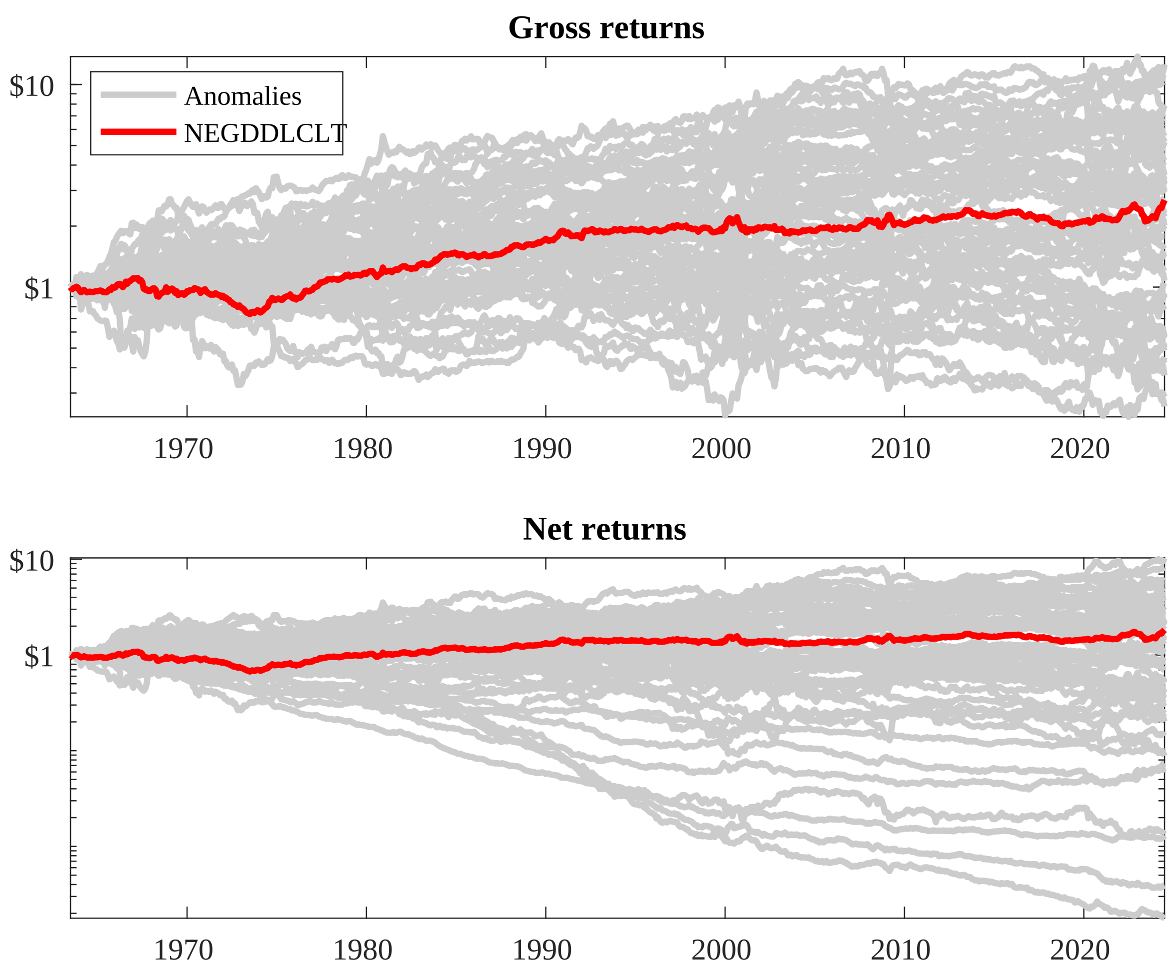
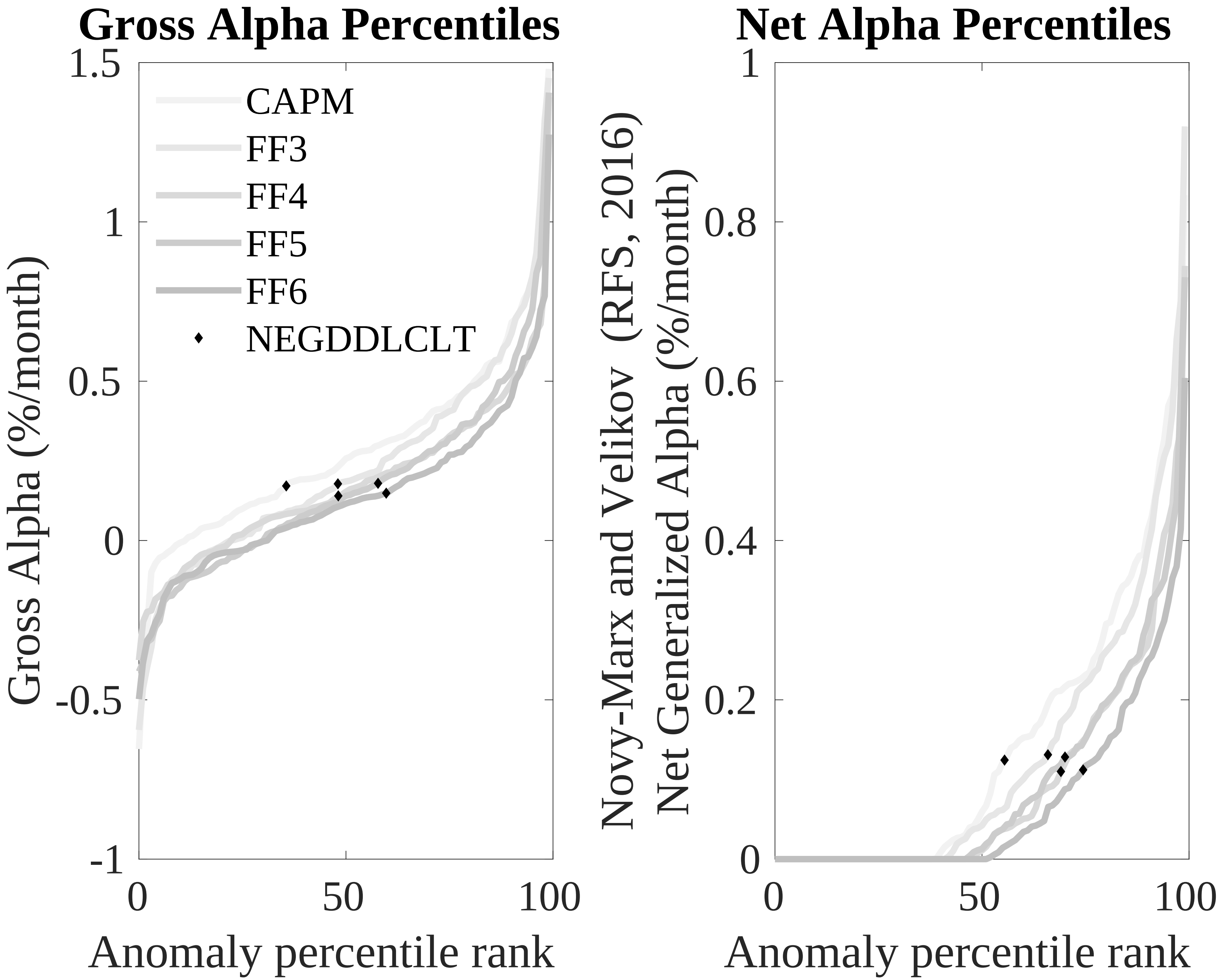
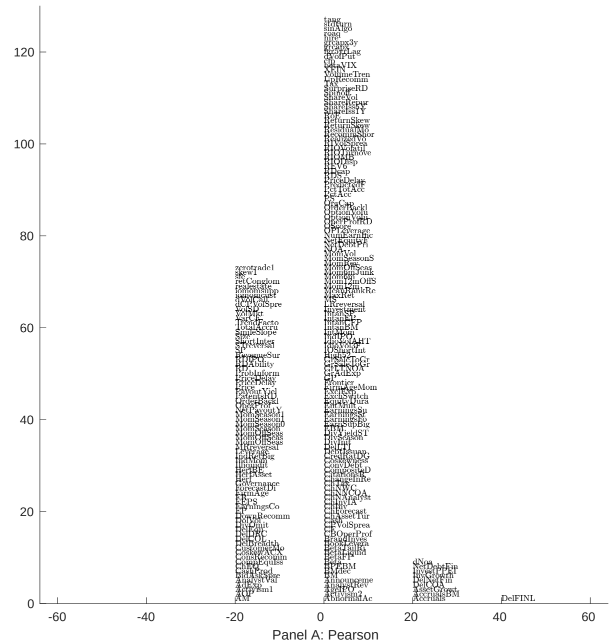


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MSS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDDLCLT
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100



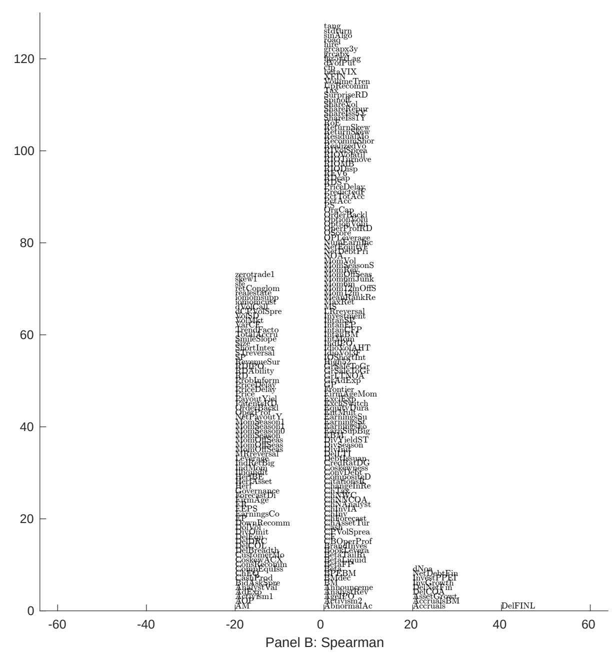


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with MSS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

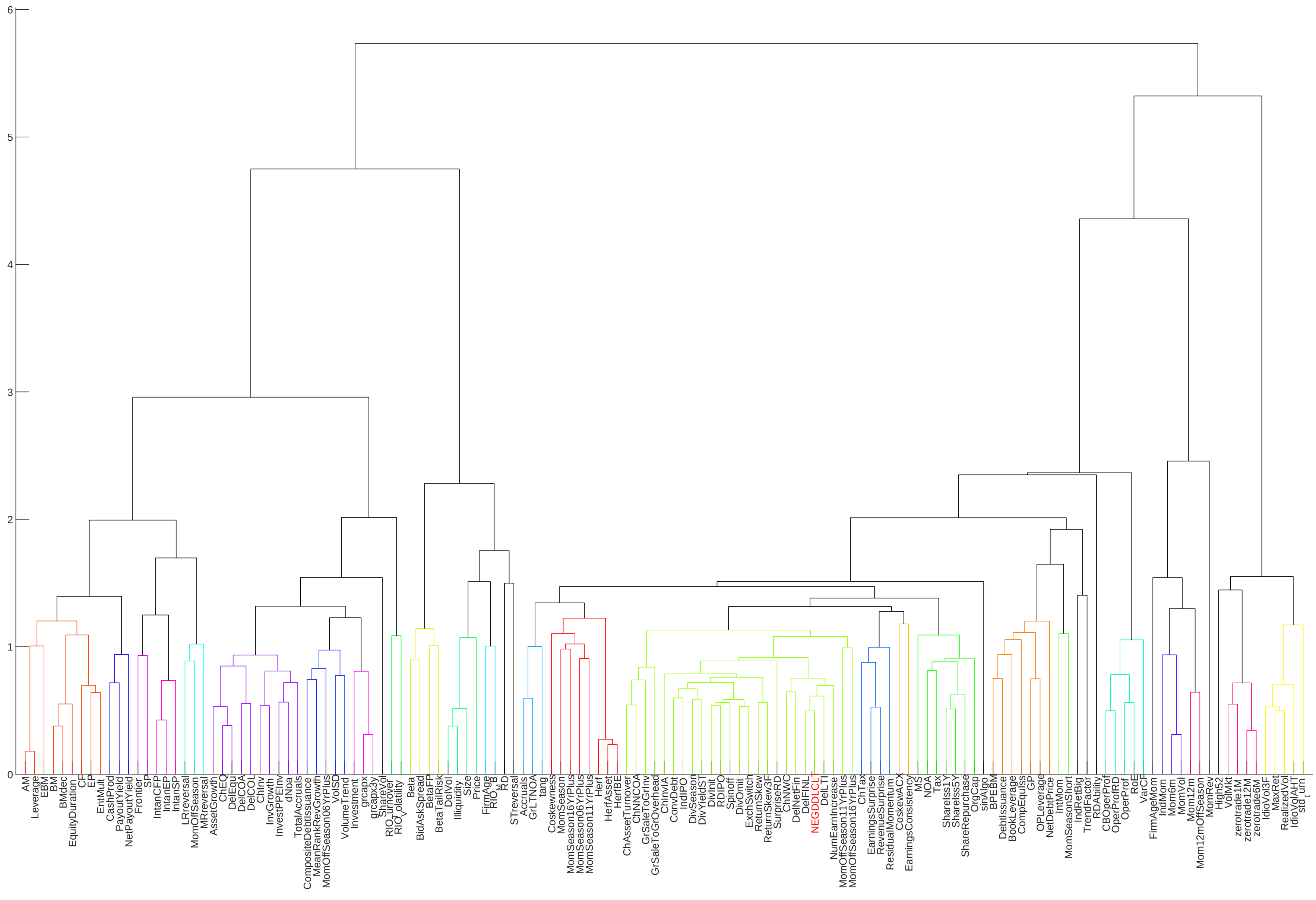


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

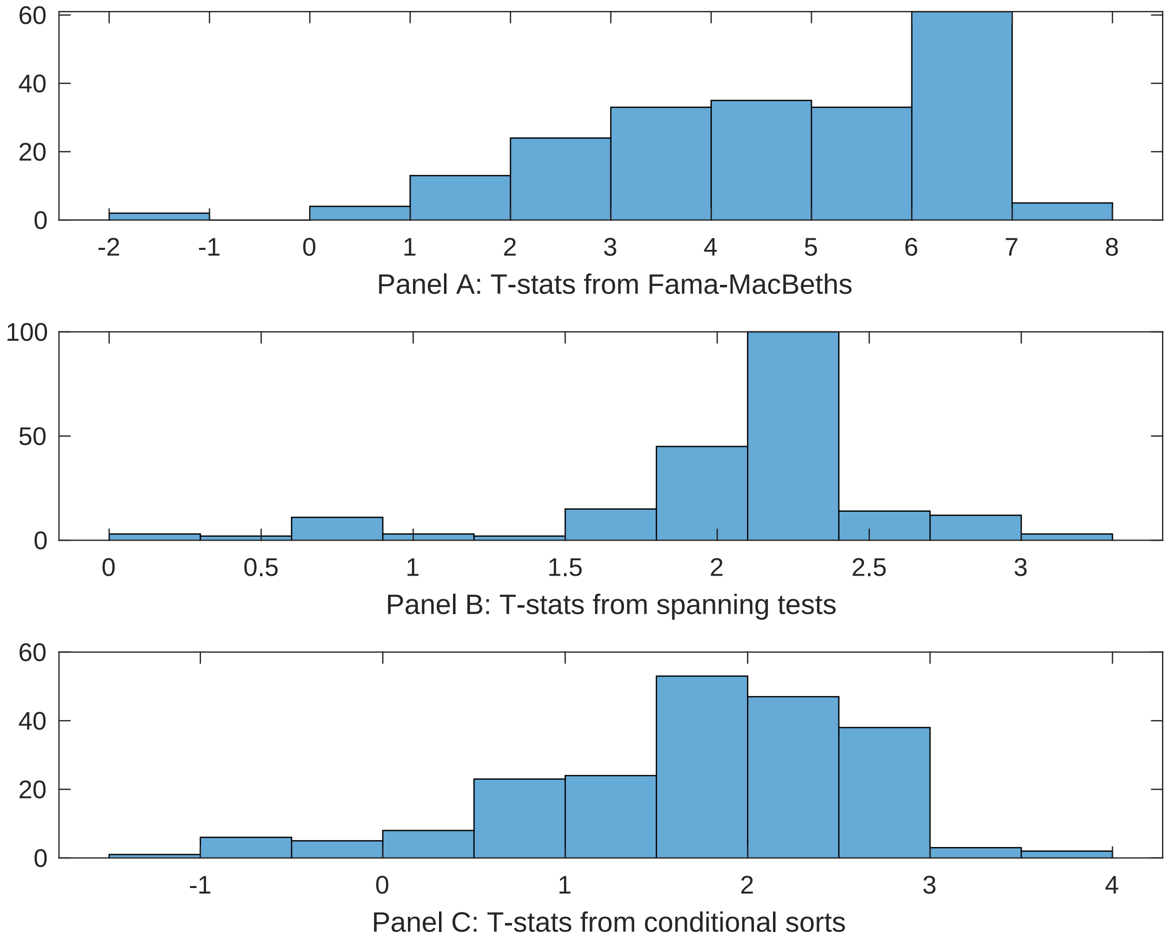


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of MSS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MSS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MSS} MSS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MSS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MSS. Stocks are finally grouped into five MSS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MSS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MSS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MSS} MSS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12	0.13	0.12	0.12	0.13	0.12	0.13
	[5.85]	[5.18]	[5.54]	[5.74]	[5.97]	[5.64]	[5.54]
MSS	0.46	0.75	0.74	0.69	0.36	0.86	0.53
	[3.19]	[4.66]	[5.19]	[5.00]	[2.40]	[6.15]	[3.13]
Anomaly 1	0.16						0.76
	[8.93]						[0.19]
Anomaly 2		0.19					0.11
		[8.53]					[3.24]
Anomaly 3			0.13				0.57
			[4.38]				[1.15]
Anomaly 4				0.74			-0.83
				[5.27]			[-3.11]
Anomaly 5					0.13		0.14
					[8.60]		[5.86]
Anomaly 6						0.60	-1.00
						[2.35]	[-2.08]
# months	732	630	732	732	720	732	630
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the MSS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MSS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.11	0.19	0.10	0.12	0.14	0.09	0.12
	[1.90]	[2.86]	[1.55]	[1.89]	[2.12]	[1.50]	[1.99]
Anomaly 1	42.34						39.01
	[12.27]						[8.03]
Anomaly 2		27.67					2.02
		[7.71]					[0.48]
Anomaly 3			18.56				10.70
			[7.24]				[4.21]
Anomaly 4				15.86			-6.46
				[5.12]			[-2.01]
Anomaly 5					15.61		-3.48
					[4.31]		[-0.94]
Anomaly 6						25.68	16.35
						[8.22]	[4.93]
mkt	-2.89	-4.08	-2.47	-3.65	-3.85	-4.18	-2.80
	[-2.03]	[-2.69]	[-1.63]	[-2.39]	[-2.50]	[-2.80]	[-2.03]
smb	-2.94	0.63	5.19	2.36	1.61	2.42	1.03
	[-1.41]	[0.27]	[2.32]	[1.06]	[0.72]	[1.12]	[0.47]
hml	-4.03	-5.61	-3.57	-7.41	-8.63	-5.46	0.71
	[-1.46]	[-1.93]	[-1.21]	[-2.50]	[-2.89]	[-1.89]	[0.26]
rmw	-12.53	-9.87	-2.61	-6.85	-8.85	-6.24	-6.17
	[-4.50]	[-3.31]	[-0.85]	[-2.27]	[-2.95]	[-2.13]	[-2.16]
cma	-3.15	1.39	7.58	15.77	1.35	11.73	-9.32
	[-0.74]	[0.31]	[1.75]	[3.60]	[0.26]	[2.77]	[-1.86]
umd	1.10	2.62	2.89	4.15	4.14	3.07	-0.27
	[0.77]	[1.72]	[1.92]	[2.73]	[2.71]	[2.06]	[-0.19]
# months	732	630	732	732	732	732	630
$\bar{R}^2$(%)	22	13	12	9	8	13	30

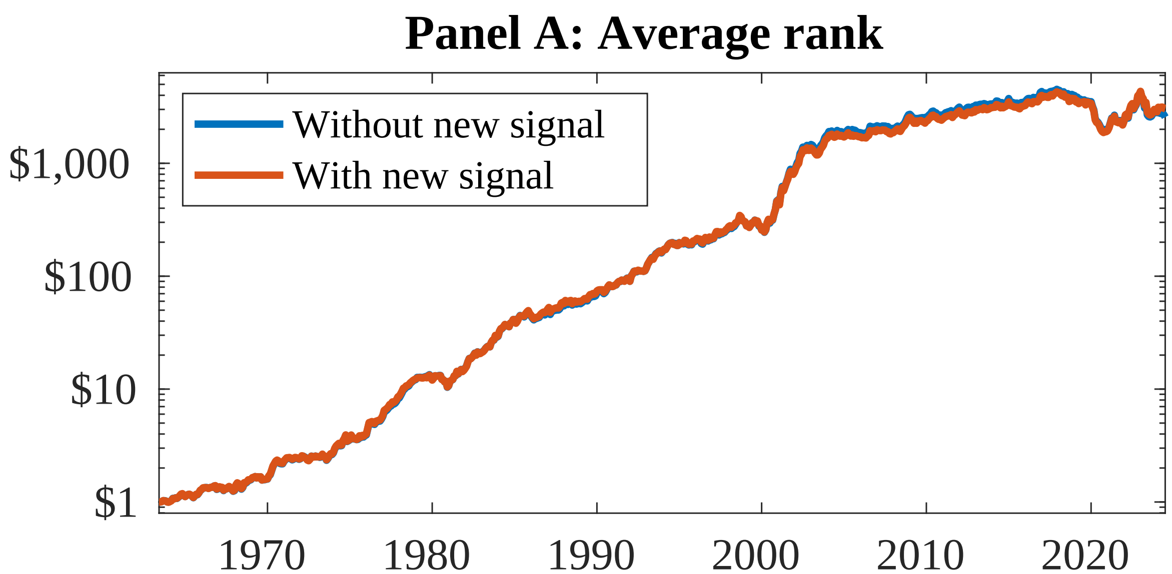


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MSS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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